# ZAIN UL ABIDIN

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Google Scholar | ORCID | GitHub | LinkedIn | ResearchGate | Portfolio

## RESEARCH PROFILE

Computer Systems Engineer with research experience in computer vision, machine learning, image processing, signal processing, Non-Destructive Testing (NDT), and Structural Health Monitoring (SHM). Co-author of eight research papers, including one published, one accepted, three under review, and three ready for submission, focused on defect detection, segmentation, thermographic imaging, depth estimation, 3D reconstruction, image super-resolution, and multimodal learning for real-world engineering applications. Actively seeking MSc/PhD research opportunities to work with professors in computer vision and AI on applied research problems in intelligent inspection and visual systems.

## RESEARCH INTERESTS

Machine Learning and Deep Learning for Industrial Applications | Computer Vision and Image Processing | Non-Destructive Testing and Evaluation | Digital Signal Processing | Structural Health Monitoring

## EDUCATION

**B.Eng. in Computer Systems Engineering (Result Awaited)** *Sep 2022 - May 2026*
*Sukkur IBA University, Sukkur, Sindh, Pakistan | iba-suk.edu.pk*

- CGPA: 3.79 / 4.00 (A = 4.0, US 4.0 GPA scale; all courses taught in English)
- PEC Accredited, Level II, Washington Accord Substantial Equivalence

## PUBLICATIONS

### Published

1. **Mitigation of DDoS Attacks with Machine Learning, Deep Learning, and Transformers** *2024*
   ***Zain Ul Abidin***, *Haseeb, Habeeban Memon, Muhammad Younis, Junaid Ahmed*
   International Journal of Multidisciplinary Conference Proceedings (IJMCP), Vol. 2(1), 2024 | DOI: 10.61503/Ijmcp.v2i1.181
   - Benchmarked ML and DL models for multi-class DDoS detection on CICDDoS2019. LightGBM and XGBoost achieved the best accuracy-speed balance for real-time network security.

### Accepted (In Press)

2. **Debond Detection in CFRP Structures using Optical Pulse Thermography and Deep Learning** *2026*
   *Habeeban Memon,* ***Zain Ul Abidin****, Junaid Ahmed, Ayesha Siddiqua*
   iCoMET 2025, IEEE Xplore | Preprint: ResearchGate
   - Compared YOLOv8, Faster R-CNN, and Mask R-CNN for CFRP debond detection. Mask R-CNN achieved the highest accuracy with pixel-level defect segmentation.

### Under Review

3. **Metal Defect Detection in Pulsed Thermographic Imaging via Temporal Eigen Projection and Robust Low-Rank Tensor Completion Model** *2025*
   ***Zain Ul Abidin***, *Junaid Ahmed, Dong Wang, Guiyun Tian*
   Nondestructive Testing and Evaluation, Taylor and Francis | Preprint: ResearchGate
   - Proposed Eigen-RHOTC combining temporal eigen projection with robust higher-order tensor completion to isolate ECPT defect features. Outperforms prior methods in SNR and processing speed.

4. **Eddy Current Pulsed Thermography based Efficient Defect Detection in Metals via Keyframe Extraction and Accelerated Sparse Decomposition** *2025*
   *Junaid Ahmed, **Zain Ul Abidin**, Guiyun Tian, Dong Wang*
   Quantitative InfraRed Thermography Journal, Taylor and Francis | Preprint: ResearchGate
   - Entropy- and density-based keyframe selection reduces thermal data volume. Accelerated CUR decomposition extracts weak defect signals from noisy backgrounds.
5. **Improving Image Super-Resolution: A Hybrid Approach Using DT-CWT and Transformer Networks** *2025*
   *Pan Zhiwen, Faisal Sahito, Aakash Kumar, Junaid Ahmed, **Zain Ul Abidin***
   Image and Vision Computing, Elsevier
   - A DT-CWT and Transformer hybrid for image super-resolution that preserves fine structural detail better than current state-of-the-art baselines.

### Ready to Submit / Preprints

6. **OPT-based Three-Channel Thermographic Fusion for Architecture-Independent Defect Detection and Segmentation in CFRP Composites** *2026*
   ***Zain Ul Abidin**, Habeeban Memon, Junaid Ahmed | Target journal to be confirmed*
   - Proposed an automated, architecture-independent input representation using a three-channel fusion based on cyclic alignment of raw thermal intensity, PCA, and TSR to isolate small, low-contrast defects.
7. **Automated Depth Estimation and 3D Reconstruction of CFRP Subsurface Defects via Optical Pulsed Thermography** *2026*
   *Habeeban Memon, **Zain Ul Abidin**, Junaid Ahmed | Target journal to be confirmed*
   - Proposed a two-stage pipeline extracting 16 physics-informed features to predict through-thickness depth and reconstruct 3D topologies of CFRP subsurface delaminations.
8. **CMM-DSE: Cross-Modal Consistency-Guided Multimodal Fake News Detection via Dempster-Shafer Evidential Reasoning** *2026*
   *Habeeban Memon, **Zain Ul Abidin**, Umair Ayaz Kamangar | Target journal to be confirmed*
   - Proposed a Cross-Modal Consistency Module and Dempster-Shafer evidential fusion layer, utilizing 2D FFT features for robust uncertainty quantification in multi-modal forgery detection.

## RESEARCH & DEVLOPMENT PROJECTS

1. **Auto-Research: Autonomous Multi-Agent Research Pipeline** *2026 – Present*
   *Multi-Agent Systems | LLM | Automated Research | Code Generation*
   - Engineering an end-to-end autonomous research pipeline capable of processing a given topic, thesis, or dataset to independently generate code, execute experiments, and validate results.
   - Designing a complex 23-stage orchestration framework utilizing specialized AI agents for planning, coding, testing, debugging, literature reviews, experiment design, and citation verification.
2. **X-ray Prohibited Item Detection** *2025 - Present*
   *Computer Vision | Deep Learning | Object Detection*
   - Developing an automated computer vision system utilizing deep learning architectures to accurately identify and segment prohibited and dangerous items within complex, highly cluttered X-ray security scan.

## ACADEMIC PROJECTS & THESIS

1. **Undergraduate Thesis | Machine Learning Based Subsurface Defect Detection in CFRP Using Optical Pulse Thermography** *May 2025 - May 2026*
   *NDT/NDE | CFRP Composites | Defect Detection | Defect Segmentation | Depth Estimation | 3d Reconstruction*

- Developed a machine learning-based system to automate the detection and analysis of subsurface defects in Carbon Fiber Reinforced Polymer (CFRP) composites.
- Processed raw optical pulsed thermography (OPT) sequences to build an end-to-end pipeline (Detection → Segmentation → Depth Estimation → 3D Reconstruction) for non-destructive testing.

2. **Smart Asset Management System (SAMS)** *2026*
   *Software Engineering | ASP.NET framework | C# | SQL Server | Web app: sams.somee.com*
   - Developed a centralized, full-stack web application framework to manage the tracking, assignment, and lifecycle of organizational assets with secure, role-based authentication.
   - Engineered an automated notification system to alert users of maintenance events, supported by a normalized relational database utilizing parameterized queries to prevent SQL injection.
3. **Real-Time Speech Emotion Recognition via LSTM** *2025*
   *Speech Processing | LSTM | Toronto Emotional Speech Set (TESS) | Repository: GitHub*
   - LSTM classifier trained on TESS with noise injection and time-shifting augmentation; achieves real-time inference on live audio streams.
4. **Gold Price Forecasting Dashboard** *2025*
   *Time-Series Forecasting | Ensemble Learning | Streamlit | Repository: GitHub*
   - Ensemble of XGBoost, LSTM, and Random Forest models trained on historical market data, visualised through an interactive Streamlit dashboard with side-by-side model comparison.
5. **Speech Enhancement via Spectral Subtraction and Wiener Filtering** *2025*
   *Signal Processing | Python | Repository: GitHub*
   - Real-time pipeline estimating background noise from degraded speech and applying Wiener filtering to recover intelligibility.
6. **Hand Gesture Recognition and Voice Conversion** *2024*
   *Computer Vision | CNN | SVM | Accessibility | Repository: GitHub*
   - CNN and SVM pipeline for sign language gesture classification; gestures converted to synthesised speech in real time for users with hearing or speech impairments.
7. **Image Compression Analysis: JPEG vs JPEG2000** *2024*
   *Image Processing | DCT | DWT | Repository: GitHub*
   - DCT-based JPEG and DWT-based JPEG2000 implemented from first principles; quality benchmarked via PSNR, SSIM, and compression ratio.
8. **AI-Based Android Controlled Vision Robot** *2024*
   *Edge AI | ESP32-CAM | TensorFlow Lite | Repository: GitHub*
   - ESP32-CAM with TensorFlow Lite for on-device real-time object detection; demonstrates edge AI deployment on low-cost embedded hardware.

## WORK EXPERIENCE

1. **Artificial Intelligence Intern** *Jul 2025 - Aug 2025*
   *Mirai School of Technology | Remote*
   - Designed and deployed a Conversational Order Assistant using agentic AI workflows (n8n, Gemini AI), automating order capture, validation, logging, and email confirmation via RESTful APIs.
2. **Software Engineering Intern** *Jun 2025 - Jul 2025*
   *Sukkur Electric Power Company (SEPCO) | On-site, Sukkur, Sindh*
   - Built the Mute Meter Monitoring Dashboard (Python, Streamlit, SQLite) with GIS zone filters and bulk import/export, reducing manual field inspection workload by an estimated 40%.

## TECHNICAL SKILLS & EXPERTISE

- **Programming & Development:** Python, MATLAB, SQL, Verilog, C#, HTML/CSS, ASP.NET Web Forms
- **Machine Learning & AI:** scikit-learn, XGBoost, LightGBM, Support Vector Regression (SVR), Random Forest
- **Deep Learning & Generative AI:** TensorFlow, PyTorch, Keras, ViT, DeBERTa-v3, BLIP, RT-DETR-L, U-Net, FPN, DeepLabV3+, LSTM, LLMs, Prompt Engineering
- **Computer Vision:** OpenCV, YOLO, Faster R-CNN, Mask R-CNN, Semantic Segmentation, Instance Segmentation
- **Explainable AI (XAI):** SHAP, Grad-CAM, LIME
- **Scientific Computing & Data Analysis:** NumPy, SciPy, Pandas, Matplotlib, Seaborn
- **Web, Automation & Software Tools:** Streamlit, n8n, Git, GitHub, Jupyter Notebook
- **Databases:** SQL Server, MySQL, SQLite, ADO.NET
- **Embedded Systems & IoT:** ESP32, Arduino, Raspberry Pi (Basic)

## HONOURS AND AWARDS

- **Prime Minister's Laptop Award** *2024*
  National merit-based award for academic excellence, Government of Pakistan.
- **Sindh Educational Endowment Fund (SEEF) Scholarship** *2022 - Present*
  Fully merit-based scholarship covering undergraduate tuition fees for the full duration of the degree.
- **Top Academic Ranking, Computer Systems Engineering** *2022 - Present*
  Consistently ranked among the highest-performing students in the cohort; CGPA 3.79 / 4.00.

## CERTIFICATIONS

- Machine Learning Specialization, DeepLearning.AI / Coursera
- Mathematics for Machine Learning and Data Science Specialization, DeepLearning.AI / Coursera
- Deep Learning Specialization: Advanced AI, Hands-on Lab, Udemy
- Large Language Models, Udemy
- Google Prompting Essentials Specialization, Google / Coursera
- Google IT Support Specialization, Google / Coursera
- Career Essentials in GitHub Professional Certificate, GitHub / LinkedIn Learning
- SQL for Developers and Data Analysts (MySQL), Udemy

## LANGUAGES

**English** (Professional) | **Urdu** (Native) | **Sindhi** (Native) | **Punjabi** (Native)

## REFERENCES

- **Reference 1:** Dr. Junaid Ahmed Bhatti, Assistant Professor, Department of Computer Systems Engineering, Sukkur IBA University, Pakistan. *FYP Supervisor & Research Co-author*. j.bhatti@iba-suk.edu.pk
- **Reference 2:** Dr.-Ing. Kashif Hussain, Assistant Professor, Department of Electrical Engineering, Sukkur IBA University, Pakistan. *Course Instructor & Project Supervisor*. kashif.memon@iba-suk.edu.pk
- **Reference 3:** Dr. Ghulam Abbas Lashari, Assistant Professor, Department of Electrical Engineering, Sukkur IBA University, Pakistan. *Course Instructor & Project Supervisor*. ghulam.abbas@iba-suk.edu.pk